Module 1: Transformer from First Principles

Unpacking a Decoder-Only LM

BRAIN team

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Lecture map

1. Next-token prediction
2. Attention from first principles
3. The Transformer block
4. From GPT-2 to modern LLMs & Lab 1

Guiding question

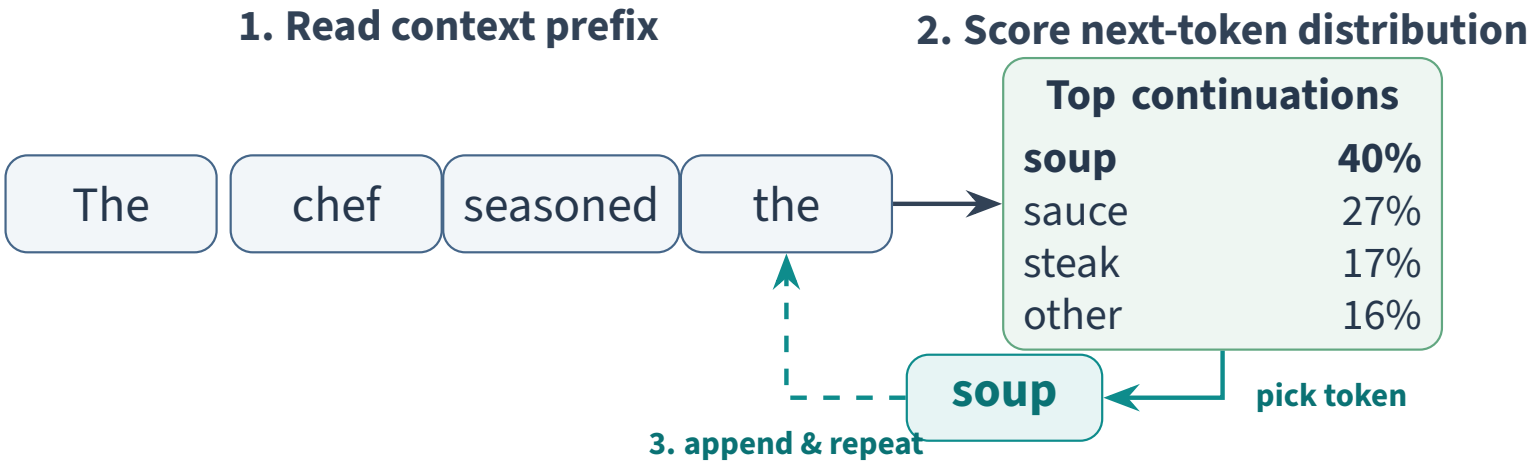
How does a decoder-only LLM generate text?

PART 1

Next-token prediction

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The task: sequence in, next-token distribution out



At every step, use the prefix to score the vocabulary $\mathcal{V}$, append one token, and repeat.

▷ **Live demo: Explore next-token prediction (next-token.html)**

Formalizing next-token prediction (NTP)

Autoregressive factorization

By the chain rule, the joint probability $P(X)$ factorizes into conditional steps:

$$P(X) = \prod_{t=1}^{T} P(x_t \mid x_{<t})$$

Modeling a text sequence is equivalent to predicting each token from prefix $x_{<t}$.

Self-supervised training loss

Natural text is self-labeled: token x_t is the target for prefix $x_{<t}$. Cross-entropy:

$$\mathcal{L}(\theta) = -\frac{1}{T} \sum_{t=1}^{T} \log P_\theta(x_t \mid x_{<t})$$

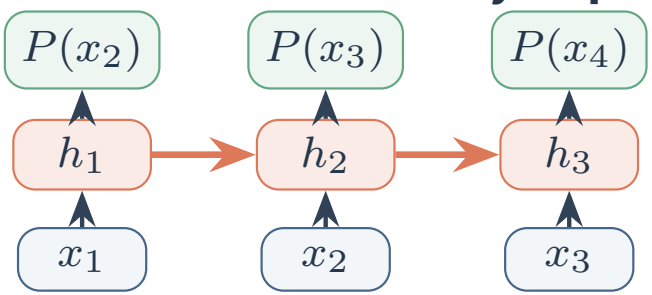
One sequence of length T provides T supervised prediction targets in parallel.

Source: Bengio et al. (2003); Radford et al. (2018, 2019)

Same task, two ways to process the sequence

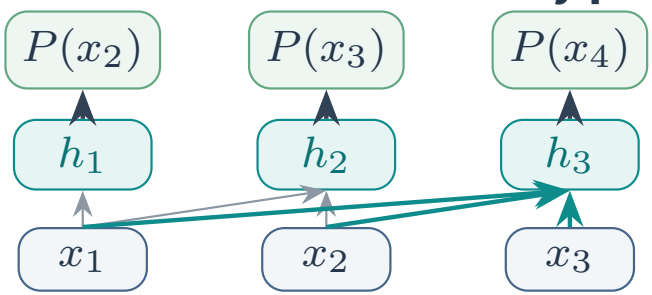
At every position t, predict $P(x_{t+1} \mid x_{\leq t})$ from the prefix so far.

Recurrent model: inherently sequential



$h_t = f(h_{t-1}, x_t)$: step t waits on step $t-1$.

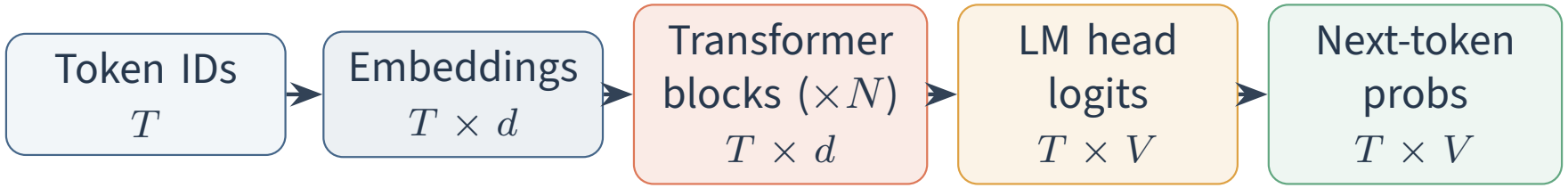
Causal Transformer: inherently parallel



No $h_{t-1} \to h_t$ wait: every position gathers $x_{\leq t}$ directly via attention in one pass.

Source: Elman (1990); Hochreiter & Schmidhuber (1997); Pascanu et al. (2013); Vaswani et al. (2017)

The whole model, at a glance



1. Embed

Token IDs index into lookup table $W_E \in \mathbb{R}^{V \times d}$ to become continuous latent vectors in $\mathbb{R}^d$.

2. Process $(\times N)$

N identical blocks alternate cross-token mixing (**attention**) and per-token reasoning (**MLP**).

3. Unembed

Linear head maps latent vectors back to vocabulary scores (logits $\in \mathbb{R}^V$); softmax yields probabilities.

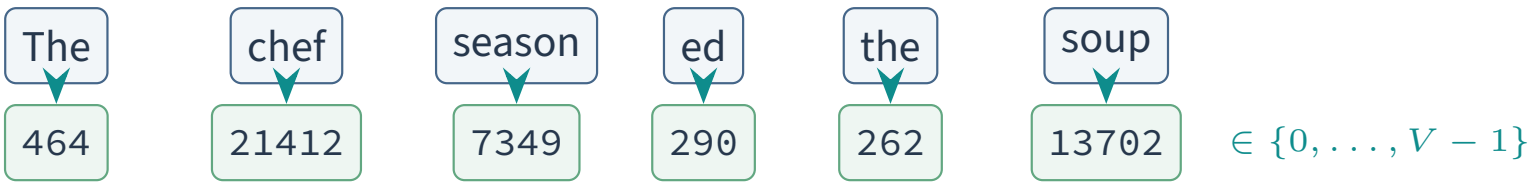
How do tokens interact inside a Transformer block?

Without recurrence h_{t-1}, each token selectively gathers information from earlier tokens in parallel via **causal self-attention**.

What is a token? From raw text to discrete integers

Text is sliced into **tokens**—atomic subword units mapped to integer IDs from a fixed vocabulary $\mathcal{V}$.

Raw text: **"The chef seasoned the soup"**



Why subwords (BPE)?

- **Optimal trade-off**: avoids T exploding (characters) and V exploding with out-of-vocabulary words.
- **Vocabulary size**: $V \approx 50\text{k}–128\text{k}$; frequent words stay intact, rare words decompose (`season` + `ed`).

Deterministic pre-processing

- **Rule-based lookup**: static trie built *before* model training; no learned parameters or gradients.
- **Rule of thumb**: in English, 1 token ≈ 0.75 words (or ≈ 4 characters of text).

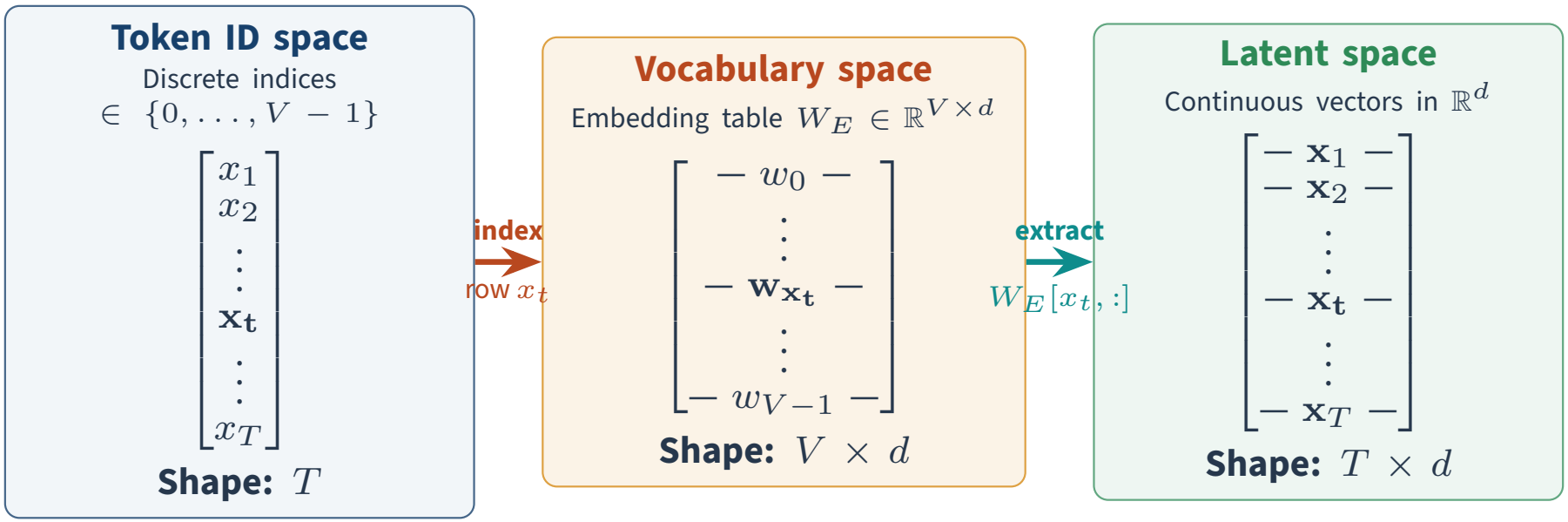
PART 2

Attention from first principles

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Step 1: Tokens become vectors in a shared space

Sequence length T · **Vocabulary size V** · **Latent dimension d**

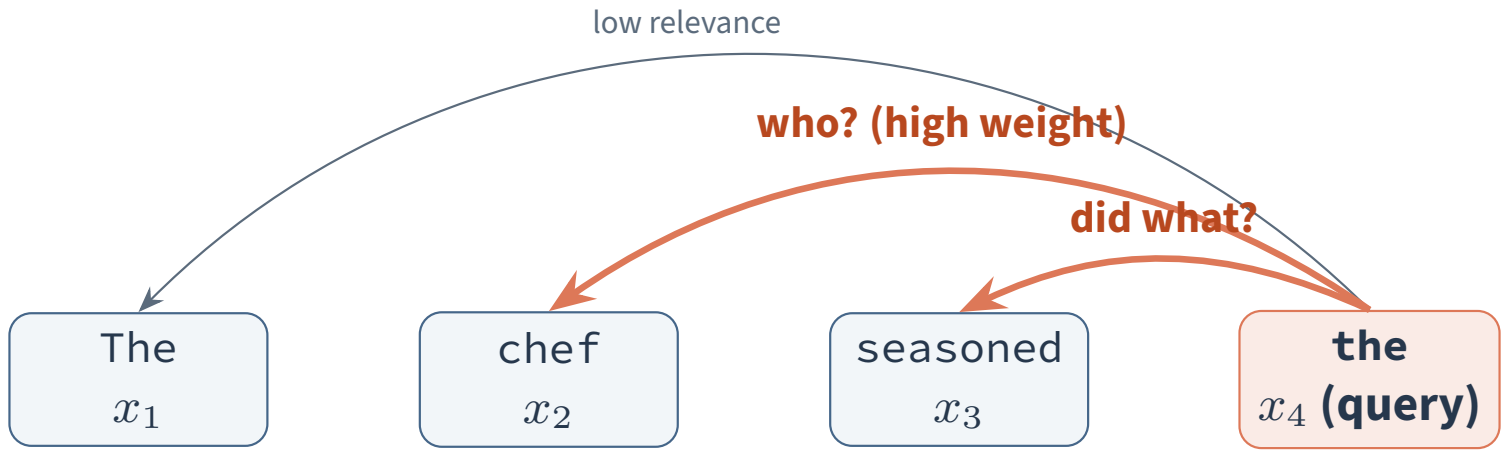


Isolated representations: context-blind vectors

Each row $\mathbf{x}_t \in \mathbb{R}^d$ purely reflects its dictionary definition ($W_E[x_t]$). At positions 1 and 4, “the” produces the exact same vector—it knows nothing yet about surrounding tokens or word order.

Step 2: Relating tokens across the sequence

To predict what follows **“the”**, its representation must pull context from related earlier tokens:



The goal of self-attention

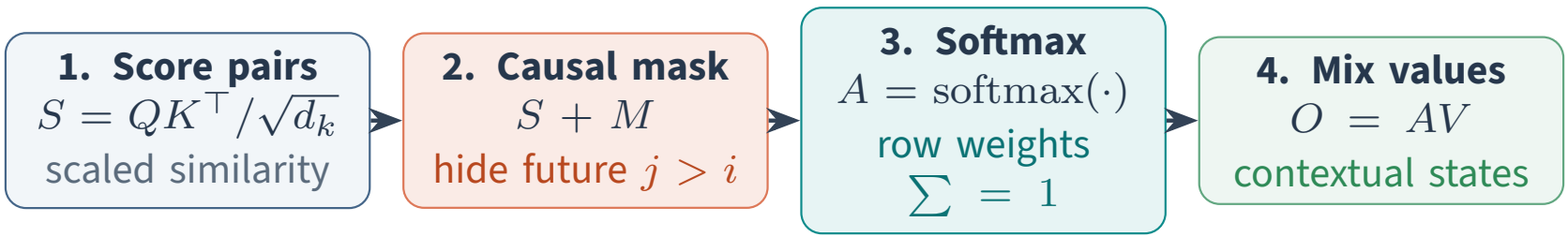
For every token i, compute a relevance weight $\alpha_{i,j} \geq 0$ ($\sum_{j \leq i} \alpha_{i,j} = 1$) for each allowed token $j \leq i$, and form a weighted mixture of their information.

Source: Bahdanau et al. (2014); Vaswani et al. (2017)

Step 3: Scaled dot-product attention

The core engine of the Transformer takes Query, Key, and Value matrices $Q, K, V \in \mathbb{R}^{T \times d_k}$:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right) V$$



Why divide by $\sqrt{d_k}$? Preserving gradient flow

For independent zero-mean unit-variance components, $\text{Var}(q^\top k) = d_k$. As dimension d_k grows, large dot products push softmax into saturated flat regions with vanishing gradients ($\nabla \approx 0$). Dividing by $\sqrt{d_k}$ keeps $\text{Var}(S) \approx 1$, maintaining healthy gradient flow.

Source: Vaswani et al. (2017, *Attention Is All You Need*)

Step 4: The intuition of Queries, Keys, and Values

Why project X into learned Q, K, V spaces instead of computing $\text{softmax}(XX^\top/\sqrt{d})X$?

Query ($q_t = x_t W_Q$)
What am I looking for?
Search criteria / prompt

Key ($k_t = x_t W_K$)
What features do I have?
Identity / catalog index

Value ($v_t = x_t W_V$)
What content do I transmit?
Content payload

Search Engine Analogy

- **Query**: search prompt entered by user.
- **Keys**: webpage titles and index tags.
- **Values**: actual page text returned.

Why separate projections?

- **Asymmetry**: $x_i^\top x_j$ is symmetric, but language relations are directed (verb $\rightarrow$ noun $\neq$ noun $\rightarrow$ verb).
- **Decoupling**: what matches a token differs from the information it delivers.

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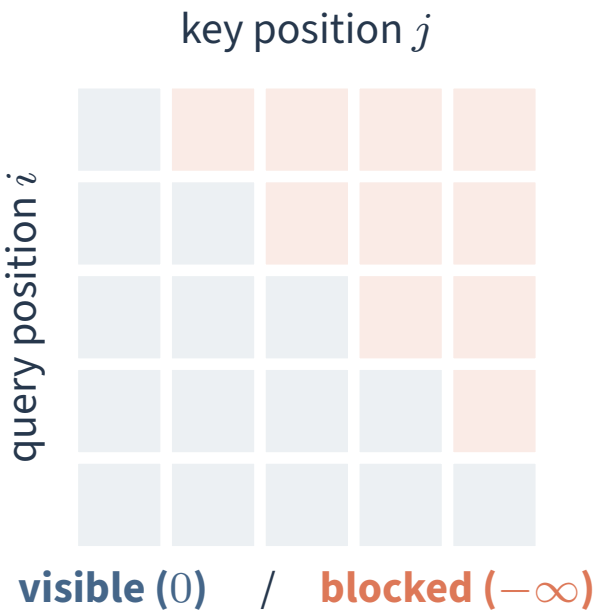
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The causal mask is a constraint on logits



- Position i may attend only to keys $j \leq i$.
- Future scores $(j > i)$ receive $-\infty$ **before** softmax so $\exp(-\infty) = 0$.
- Zeroing probabilities after softmax breaks row normalization.

Checkpoint: one query row

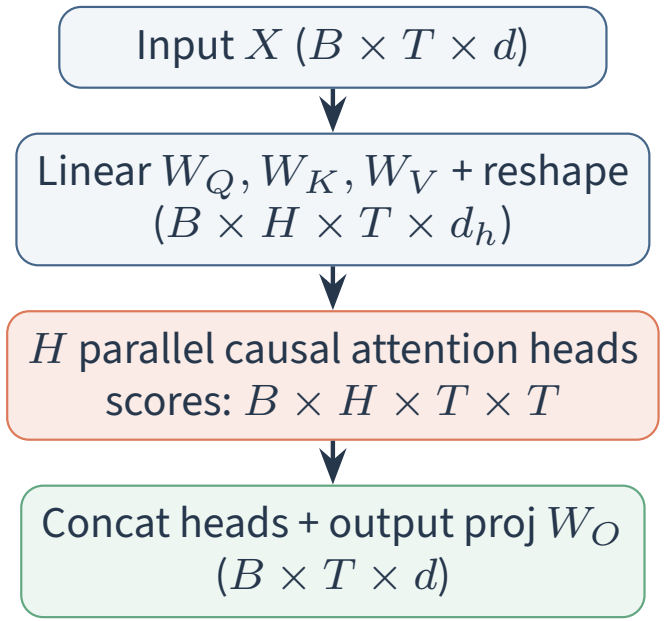
Think, then compare

For sequence length four, write the four logits seen by the query at position two (0-indexed). Which entries survive the mask, and across which dimension must softmax normalize?

Multi-head attention: multiple parallel retrieval subspaces

One token needs multiple types of context at once (syntax, coreference, local phrases). We split width d into H heads of size $d_h = d/H$:

1. **Project**: $Q, K, V \in \mathbb{R}^{B\times T\times d}$ via $W_Q, W_K, W_V \in \mathbb{R}^{d\times d}$.
2. **Split heads**: `view(B, T, H, d_h).transpose(1, 2)` $\rightarrow B \times H \times T \times d_h$.
3. **Attend per head**: $S = QK^\top/\sqrt{d_h} \in \mathbb{R}^{B\times H\times T\times T}$, mix $O_h = AV_h$.
4. **Merge & project**: `transpose(1, 2).view(B, T, d)` then multiply by $W_O \in \mathbb{R}^{d\times d}$.



Transpose trap

Always `transpose(1, 2)` between T and H so each head sees a full $T \times d_h$ sequence!

Source: Vaswani et al. (2017)

Wait—attention alone is blind to token order!

Permutation problem

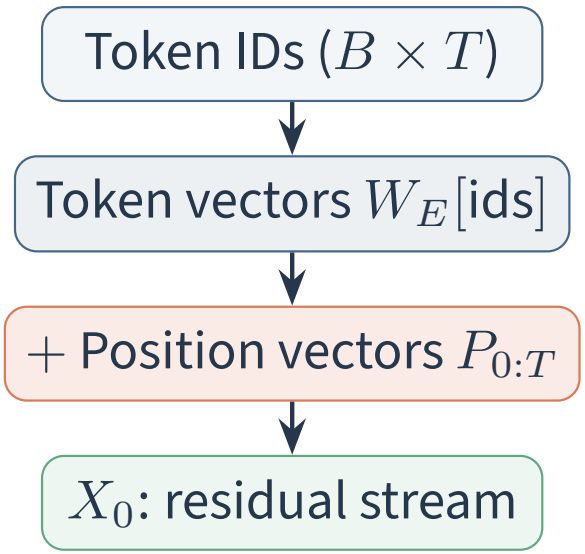
Dot products $q_i^\top k_j$ depend only on token vectors, not their indices. Without positions, attention treats the prefix like an unordered **bag of words**.

Solution: Positional Encoding (PE)

Inject position information so identical words at different positions have distinct vectors:

$$X_0 = W_E[\text{ids}] + P_{0:T} \quad \in \mathbb{R}^{B\times T\times d}$$

GPT-2: learned $W_{pe} \in \mathbb{R}^{T_{\max}\times d}$. **Modern LLMs**: RoPE rotates Q, K by relative distance.



Source: Vaswani et al. (2017); Radford et al. (2019); Su et al. (2021, RoPE)

PART 3

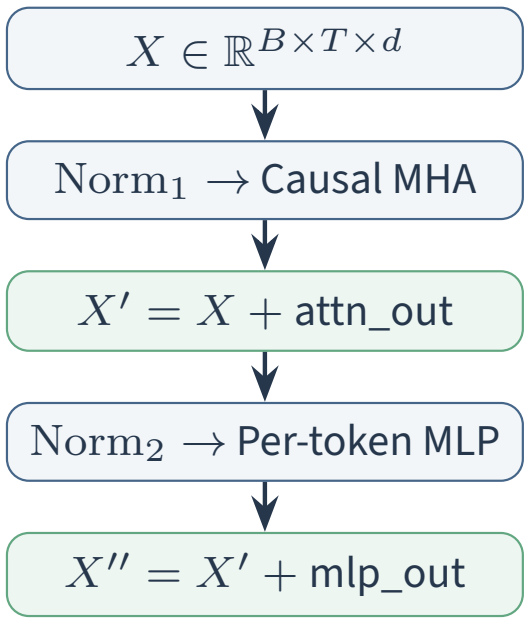
The Transformer block

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A pre-norm Transformer block has two residual updates

$$X' = X + \mathrm{MHA}(\mathrm{Norm}_1(X))$$
$$X'' = X' + \mathrm{MLP}(\mathrm{Norm}_2(X'))$$

- **Self-attention** routes information **across positions** (sequence axis T).
- **The MLP** transforms features **within each position** independently (channel axis d).
- **Residual highway** ($X + \dots$) lets gradients flow unimpeded to early layers.



Source: He et al. (2016, ResNet); Radford et al. (2019); Xiong et al. (2020)

Inside the MLP and the $1/3$ vs. $2/3$ parameter rule

The MLP applies a two-layer feed-forward network to each token vector $x_t \in \mathbb{R}^d$ independently:

$$\mathrm{MLP}(x_t) = \mathrm{GELU}(x_t W_1 + b_1) W_2 + b_2$$

- **Expand**: $W_1 \in \mathbb{R}^{d \times 4d}$ projects $d \to 4d$.
- **Activate**: elementwise GELU non-linearity.
- **Contract**: $W_2 \in \mathbb{R}^{4d \times d}$ projects $4d \to d$.

Parameter budget per block

- **MHA** (W_Q, W_K, W_V, W_O): $\mathbf{4d^2}$ **(1/3)**
- **MLP** (W_1, W_2): $\mathbf{8d^2}$ **(2/3)**
- **Total per block**: $\mathbf{12d^2}$ parameters

Takeaway for scaling

Two-thirds of block parameters and matrix-multiply FLOPs live in the MLP!

Source: Vaswani et al. (2017); Hendrycks & Gimpel (2016, GELU)

Normalization: stabilizing activations and gradients

LayerNorm (per token across d)

For each token vector $x \in \mathbb{R}^d$:

$$\mathrm{LN}(x) = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} \odot \gamma + \beta$$

Centers to zero mean (μ), scales to unit variance (σ^2), then applies learned $\gamma, \beta \in \mathbb{R}^d$.

Pre-Norm vs. Post-Norm

- **Pre-Norm (GPT-2 / modern LLMs)**: $X + F(\mathrm{Norm}(X))$. Identity highway bypasses norm; stable gradients at initialization.
- **Post-Norm (2017 Transformer)**: $\mathrm{Norm}(X + F(X))$. Requires careful warmup; different model!

Modern speedup: RMSNorm drops mean-centering (μ) and β, normalizing only by $\sqrt{\frac{1}{d}\sum_i x_i^2 + \epsilon}$.

Source: Ba et al. (2016, LayerNorm); Zhang & Sennrich (2019, RMSNorm); Xiong et al. (2020)

Closing the loop: final norm, LM head, and target shift

After N Transformer blocks, normalize X_N and project to vocabulary logits:

$$Z = \mathrm{Norm}_f(X_N) W_{\text{vocab}}^{\top} \qquad \in \mathbb{R}^{B \times T \times V}$$

$$\mathcal{L} = -\frac{1}{BT} \sum_{b,t} \log \mathrm{softmax}(Z_{b,t})_{y_{b,t}}$$

- **Weight tying (GPT-2)**: shares $W_{\text{vocab}} = W_E \in \mathbb{R}^{V \times d}$, saving $V \times d$ parameters (≈ 38.6M in GPT-2 Small).

One stream, two views

```
x = tokens[:, :-1]
y = tokens[:, 1:]
```

Common bug

Forgetting the $+1$ shift trains the model to copy the current token ($x_t \to x_t$): loss drops to zero, generation fails completely.

Source: Press & Wolf (2017, Weight Tying); Radford et al. (2019)

Interactive walkthrough: visual explorer of the Transformer

Session cue — trace a live forward pass:

1. Follow **token + position embeddings** into the residual stream.
2. Zoom into the **Transformer block**: inspect Q, K, V projections, causal attention weights, and head concatenation.
3. Trace the **MLP expansion/contraction** and final **LM head** to next-token logits.

Visual Explorer Demo

Open the interactive canvas and hover over each component to inspect tensor shapes and data flow:

▷ **Open decoder-transformer.html**

See also: Transformer Explainer (GPT-2 live)

PART 4

From GPT-2 to modern LLMs & Lab 1

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Three Transformer families—and why Decoder-Only won

Encoder-only

BERT (Devlin et al., 2018)

Bidirectional attention for classification and search. No autoregressive generation.

Encoder–decoder

Original / T5 (Raffel, 2020)

Bidirectional source encoder + causal target decoder via cross-attention.

Decoder-only

GPT / LLaMA (Radford, 2019)

Single causal stream unifies all tasks as prompt completion + single KV cache.

Why decoder-only dominates modern foundation models

Any task can be framed as prompt continuation, concentrating all parameters and pretraining compute into one unified causal stack with a single KV cache.

Source: Vaswani et al. (2017); Devlin et al. (2018); Radford et al. (2019); Brown et al. (2020)

What changed since GPT-2? The modern LLM stack

You will implement **GPT-2** in Lab 1. Modern LLMs keep the same macro-architecture while upgrading five components:

Component	GPT-2 (2019)	Modern LLM (2023+)	Primary rationale
Positions	Learned W_{pe}	**RoPE** (Su et al., 2021)	Relative distance, length extrapolation
Normalization	Pre-**LayerNorm**	Pre-**RMSNorm** (Zhang, 2019)	Drops mean-centering; 10–50% faster
Attention	**MHA** ($H_Q = H_{KV}$)	**GQA** (Ainslie et al., 2023)	$4\times$–$8\times$ smaller KV cache in serving
MLP activation	**GELU** ($d \to 4d \to d$)	**SwiGLU** (Shazeer, 2020)	Gated linear unit improves capacity
Biases & tying	Biases; tied W_E	**Bias-free**; untied W_{vocab}	Faster GEMMs; better stability at scale

Source: Touvron et al. (2023, LLaMA); Jiang et al. (2023, Mistral); Raschka (LLM Architecture Gallery)

A shape ledger catches semantic bugs

Tensor	Shape	Role in `nanolm/model.py`
`idx`	$B \times T$	input token IDs ($T \leq$ block_size)
`x`	$B \times T \times d$	residual stream (token + position)
`q, k, v`	$B \times H \times T \times d_h$	per-head projections ($d_h = d/H$)
`att`	$B \times H \times T \times T$	masked attention weights (rows sum to 1)
`mlp_hidden`	$B \times T \times 4d$	expanded per-token MLP features
`logits`	$B \times T \times V$	next-token vocabulary scores

Review question

Over which axis of `att` ($B \times H \times T \times T$) does `softmax` normalize, and why must masking happen before it?

Lab 1: what you have the tools to build and verify now

Implement in `nanolm/model.py`

1. **`CausalSelfAttention`**: W_Q, W_K, W_V, head split, $QK^\top/\sqrt{d_h}$, causal buffer mask, W_O.
2. **`MLP` & `TransformerBlock`**: $d \to 4d \to \mathrm{GELU} \to d$ and pre-LN residual updates.
3. **`MiniGPT`**: $W_E + W_{pe}$, N blocks, final LN, tied `lm_head`, cross-entropy loss.

Three verification gates

1. **Shape tests**: all intermediates match the ledger.
2. **Causality tests**: mutating a future token leaves earlier logits identical.
3. **Official GPT-2 parity**: load 124M weights (`01_gpt2_parity.py`) and match HF logits on `"The capital of France is"` → `" Paris"`.

`uv run pytest tests/test_module_01_model.py -q` → `uv run --extra gpt2 python scripts/01_gpt2_parity.py`

Checkpoint: debugging order

Think, then compare

Your model passes all shape tests, but `01_gpt2_parity.py` produces wrong logits and the causality test fails. Which lines of `CausalSelfAttention.forward` do you inspect first?

Key takeaways

1. Next-token prediction factorizes sequence probability; removing the $h_{t-1} \to h_t$ dependency makes training inherently parallel across positions on GPUs.
2. Causal multi-head attention routes information across tokens ($1/3$ of params), while positional encodings supply token order.
3. A pre-norm Transformer block pairs cross-token attention with a per-token $4d$ MLP ($2/3$ of params) along a clean residual highway.
4. Shape invariants, causality perturbation tests, and exact GPT-2 checkpoint parity verify the model before we train it in Module 2.